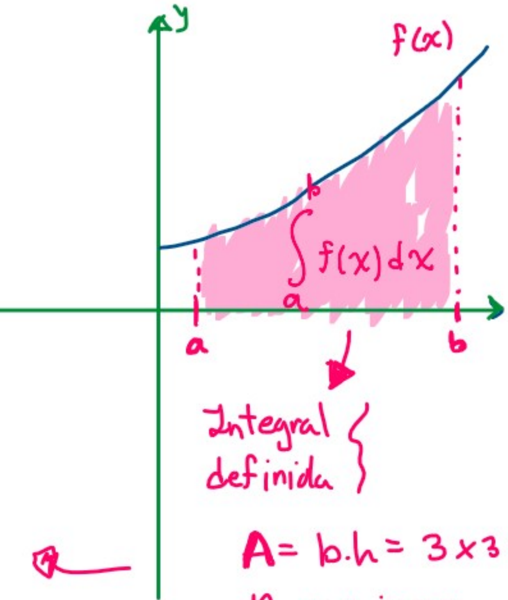
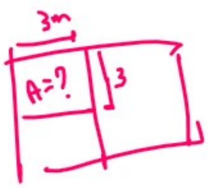
Área bajo la curva



Integral definida

exponencial

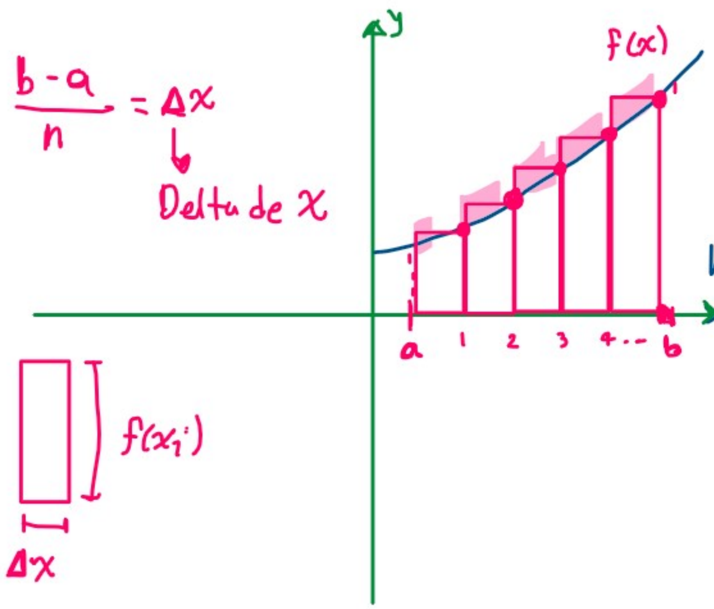
$\int f(x)\,dx$ → Integrales indefinidas



$A = b.h = 3 \times 3 = 9\,m^2$

$A \to$ número

$\frac{b-a}{n} = \Delta x$ → Delta de x



$n = 5$ rect.

Área del rectangulo $= \Delta x \cdot f(x_i)$

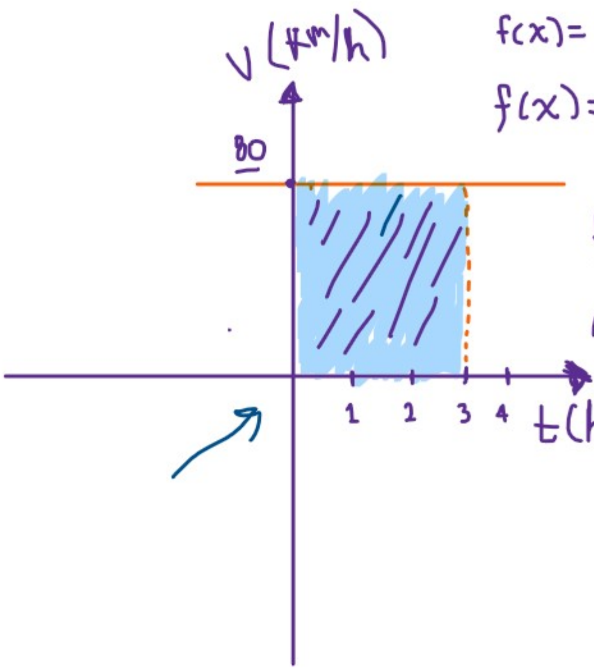
$$\tilde{A} = \sum_{i=1}^{n} f(x_i) \cdot \Delta x$$ → área aproximada

$$\lim_{n \to \infty} \sum_{i=1}^{n} f(x_i) \cdot \Delta x$$ } Integrales !!

$$\lim_{n\to\infty} \sum_{i=1} f(x_i)\cdot\Delta x \quad \Big\} \quad \text{Integrales} \ldots$$

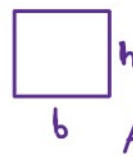
Suma de Riemann

$$\int_a^b f(x)\,dx = \boxed{\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\cdot\Delta x} \quad \Big\} \quad \text{Notación}$$



$f(x) = C$

$f(x) = 80$



$A = b \times h$

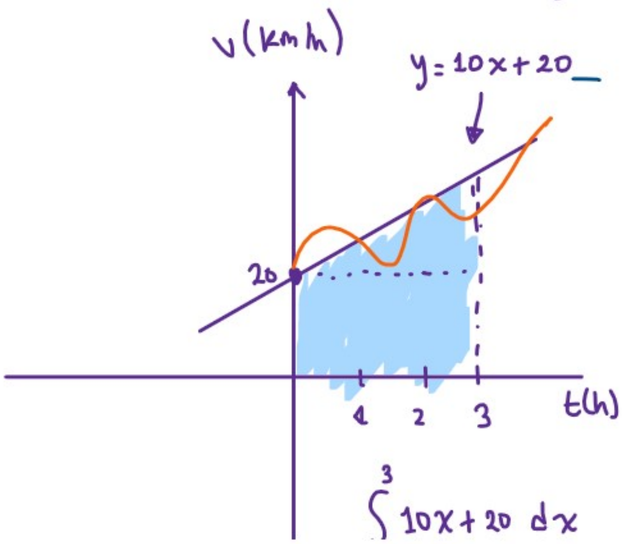
$b = 3\,h$

$h = 80\ \text{Km/h}$

$A = 3\cancel{h} \cdot 80 \frac{\text{Km}}{\cancel{h}} = 240\ \text{km}$

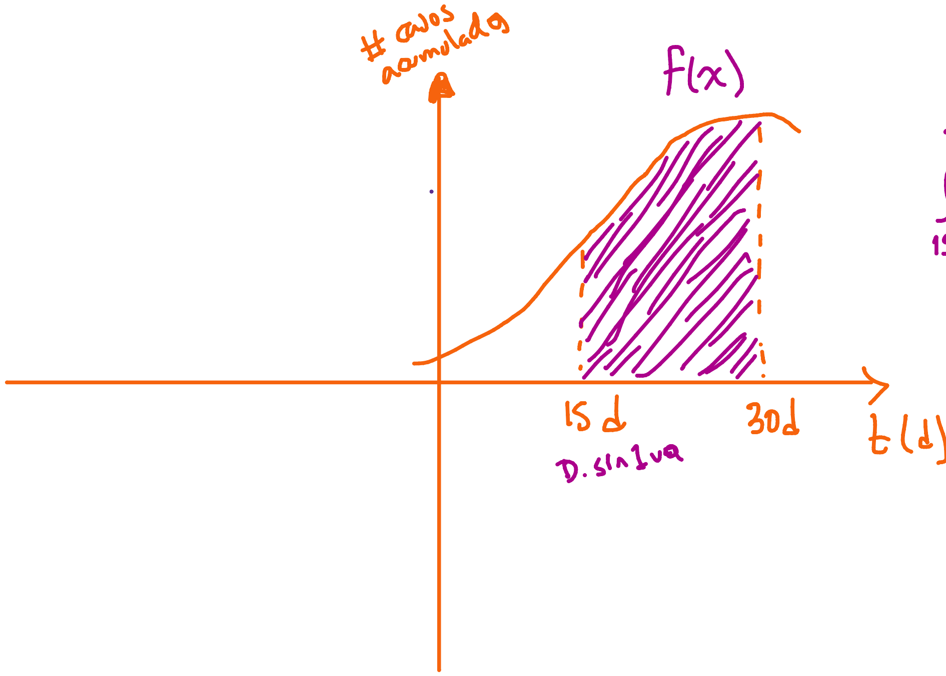
Distancia

Lineal $y = \underline{m}x + \underline{b}$



$$\int^{3} 10x + 20\ dx$$

$$\int_0^3 10x+20\,dx$$



$$\int_{15}^{30} f(x)\,dx$$